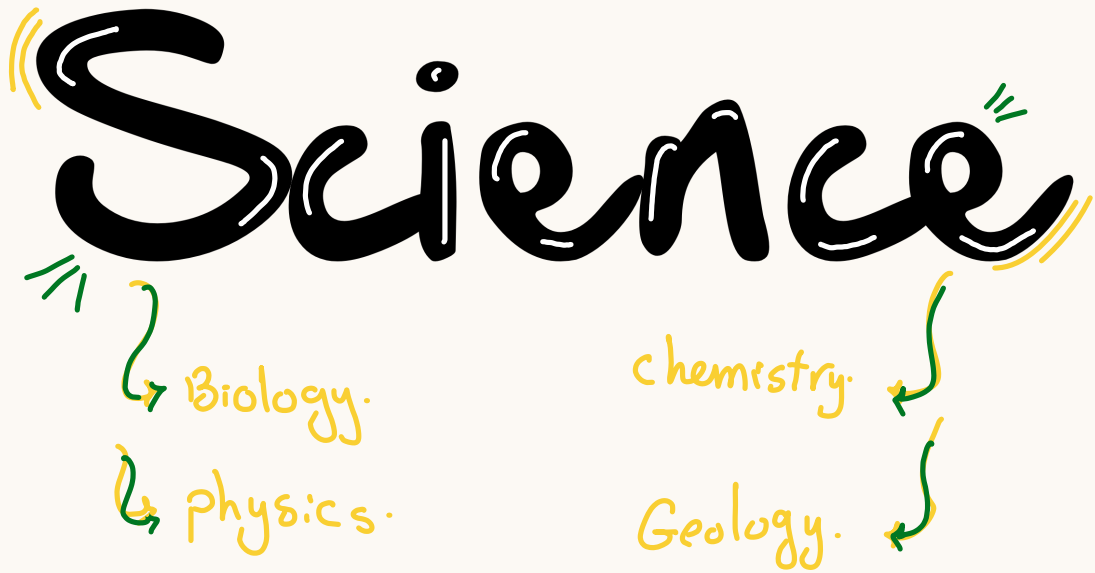


ازيكم يا شباب انا مريم محمد senior 2b في Stem maadi و ده ملخص
عملته لما كنت زيكم و لسا بقدم ف هو لامم اهم المعلومات عشان
تعرفو تحلو ف عايزه منكم دعوة حلوه عشان انا هبقا تالته و لكم المثل
من الدعاء و ربنا يوفقكم يارب



[Final position - Initial p]

Rules

① speed = $\frac{\text{Distance (meter)}}{\text{time (sec)}}$ → km/h
 ↳ scalar quantity. ↳ m/s

② velocity = $\frac{\text{Displacement}}{\text{time}}$
 ↳ vector quantity → has magnitude & direction.

③ Force = Mass × Acceleration
 ↳ gravity of the planet.
 Newton (N) kg
 • earth → 9.8 or 10 m/s²

④ Work = Force × Distance
 ↳ meter
 Joules (J)

⑤ Density = $\frac{\text{Mass}}{\text{Volume}}$
 ↳ gram or kg
 ↳ cm³ or L
 g/cm³ kg/L

⑥ Weight = Mass × Gravity
 ↳ kg ↳ 9.8 or 10
 Newton (N)

⑦ potential energy = Mass × Gravity × height
 ↳ weight ↳ Meter
 Joules (J)

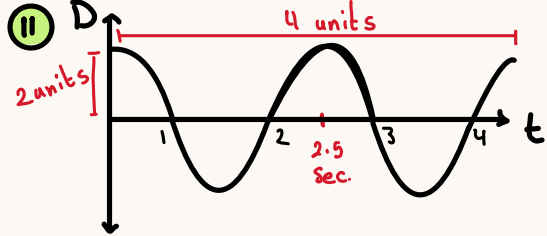
⑧ Kinitic Energy = $\frac{1}{2} \text{Mass} \times (\text{velocity})^2$
 ↳ kg ↳ (m/s)²
 Joules

⑨ Mechanical = Kinitic + Potential Energy

⑩ Relative speed:

1- $V_1 - V_2$ → same direction

2- $V_1 + V_2$ → opposite direction



• periodic time = $\frac{\text{time}}{\text{no. of vibration}}$
 ↳ sec.

• Frequency = $\frac{\text{no. of vibration}}{\text{time}} = \frac{1}{\text{periodic time}}$

• wave length (λ) = $\frac{\text{total distance}}{\text{no. of waves}}$

• wave velocity = Frequency × wave length

⑫ Acceleration of speed = $\frac{V_2 - V_1}{t}$

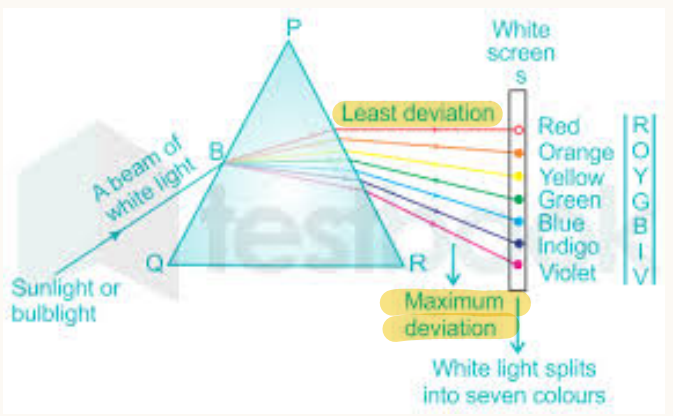
⑬ Laws of light reflection.

1. angle of reflection = angle of incident

2. incident light ray, reflected light ray and the normal line all lie on one plane which is perpendicular to reflecting surface.

⑭ Focal length = $\frac{1}{2} \times \text{radius of (R) mirror.}$

- ⑮ Intensity = $\frac{\text{quantity of } e^-}{\text{time}}$
(I) (Ampere) Coulomb
- ⑯ potential difference = $\frac{\text{work}}{\text{quantity of } e^-}$
(Volt)
- ⑰ resistance = $\frac{\text{potential diff.}}{\text{Intensity}}$
(ohm)
- ⑱ Force $\times$ arm distance = Resistance $\times$ arm distance
- ⑲ Gravity law $\rightarrow F = \frac{G M_1 M_2}{d^2}$
savart's wheel. constant
- ⑳ Sound frequency = $\frac{\text{number of times} \times \text{number of teeth}}{\text{time (sec)}}$



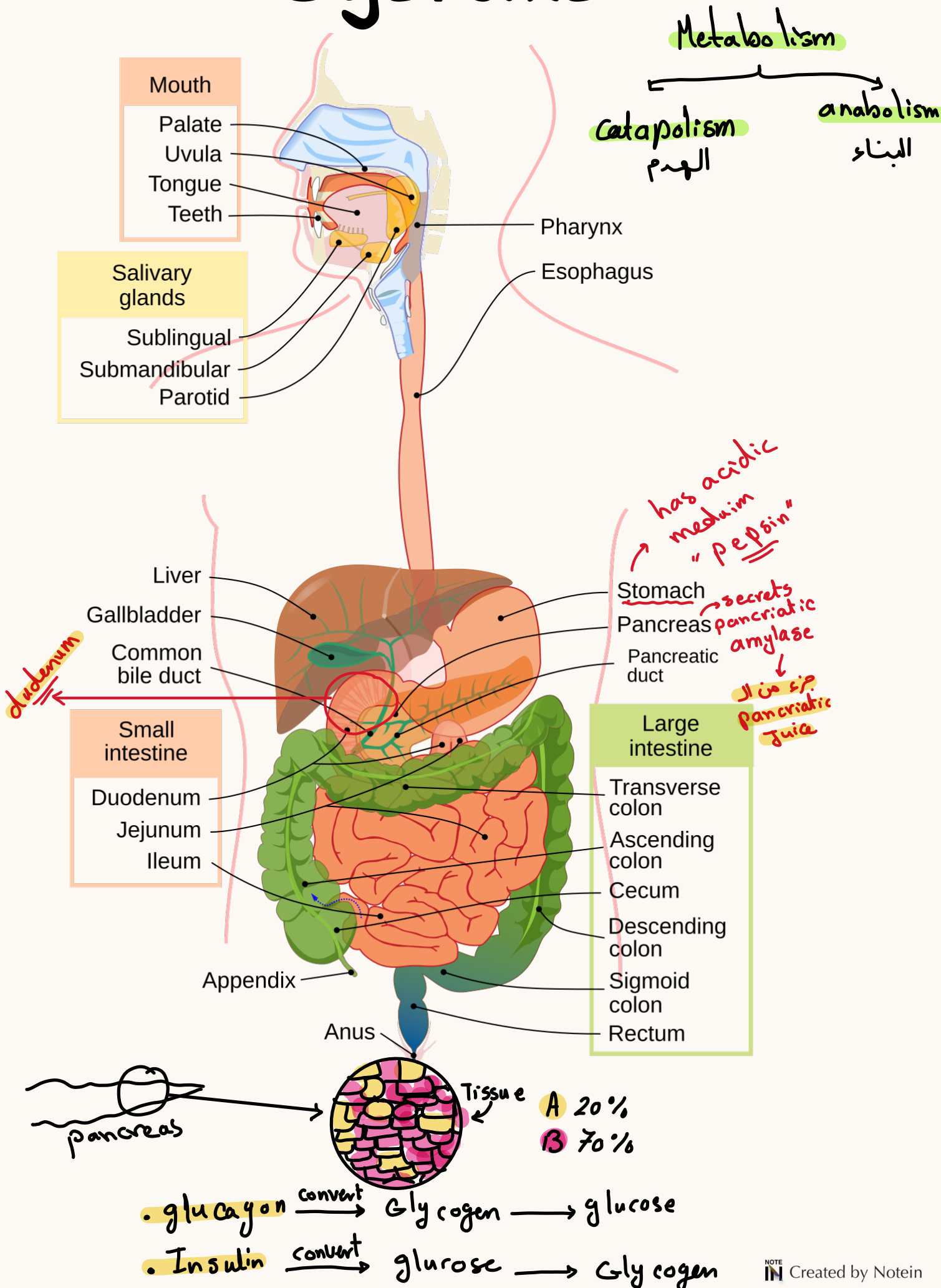
Relations

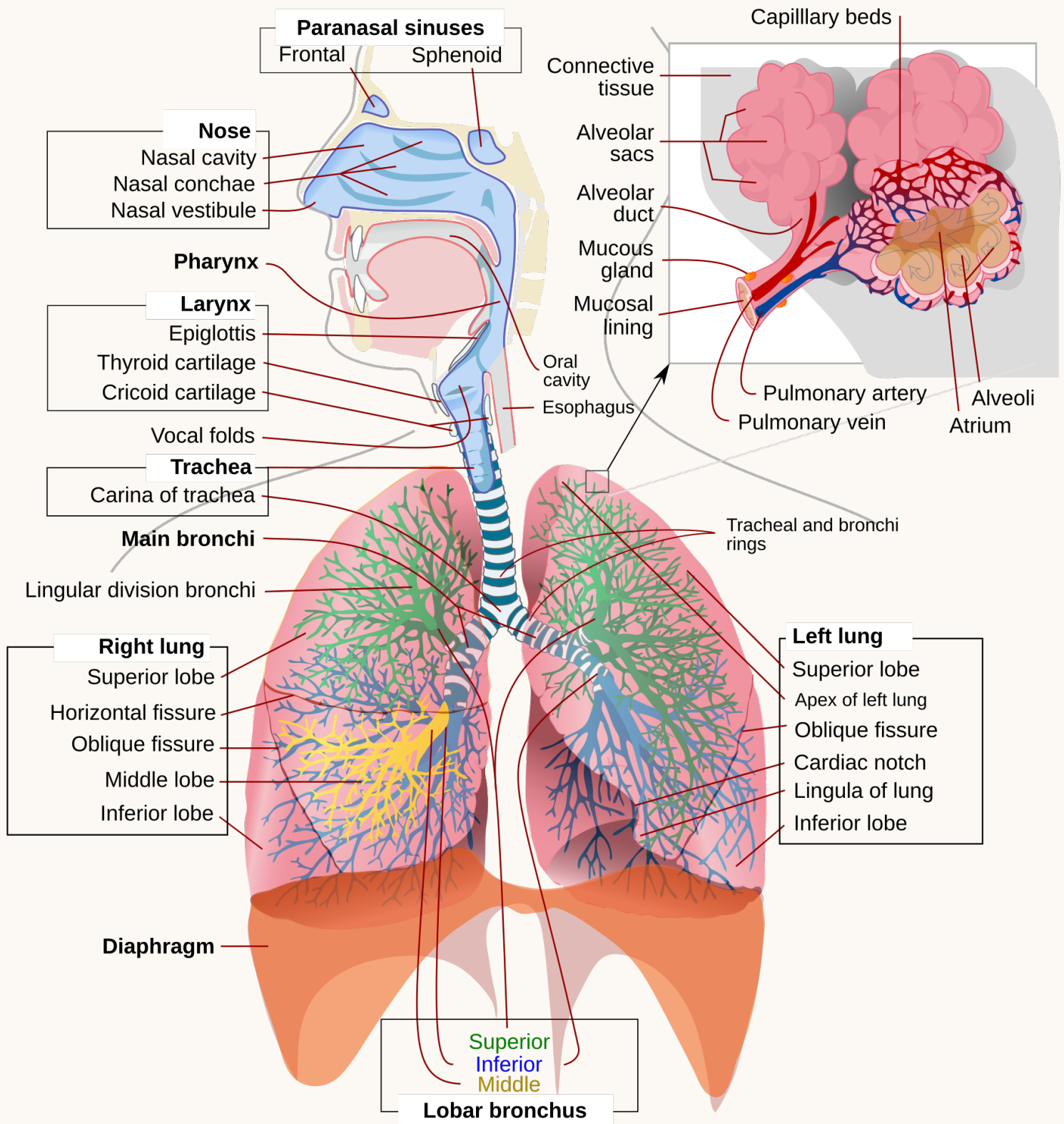
- ① Sound pitch $\propto$ Frequency "direct relation"
- ② air column $\propto \frac{1}{\text{Frequency}}$ "inverse"
- air column $\propto \frac{1}{\text{pitch}}$

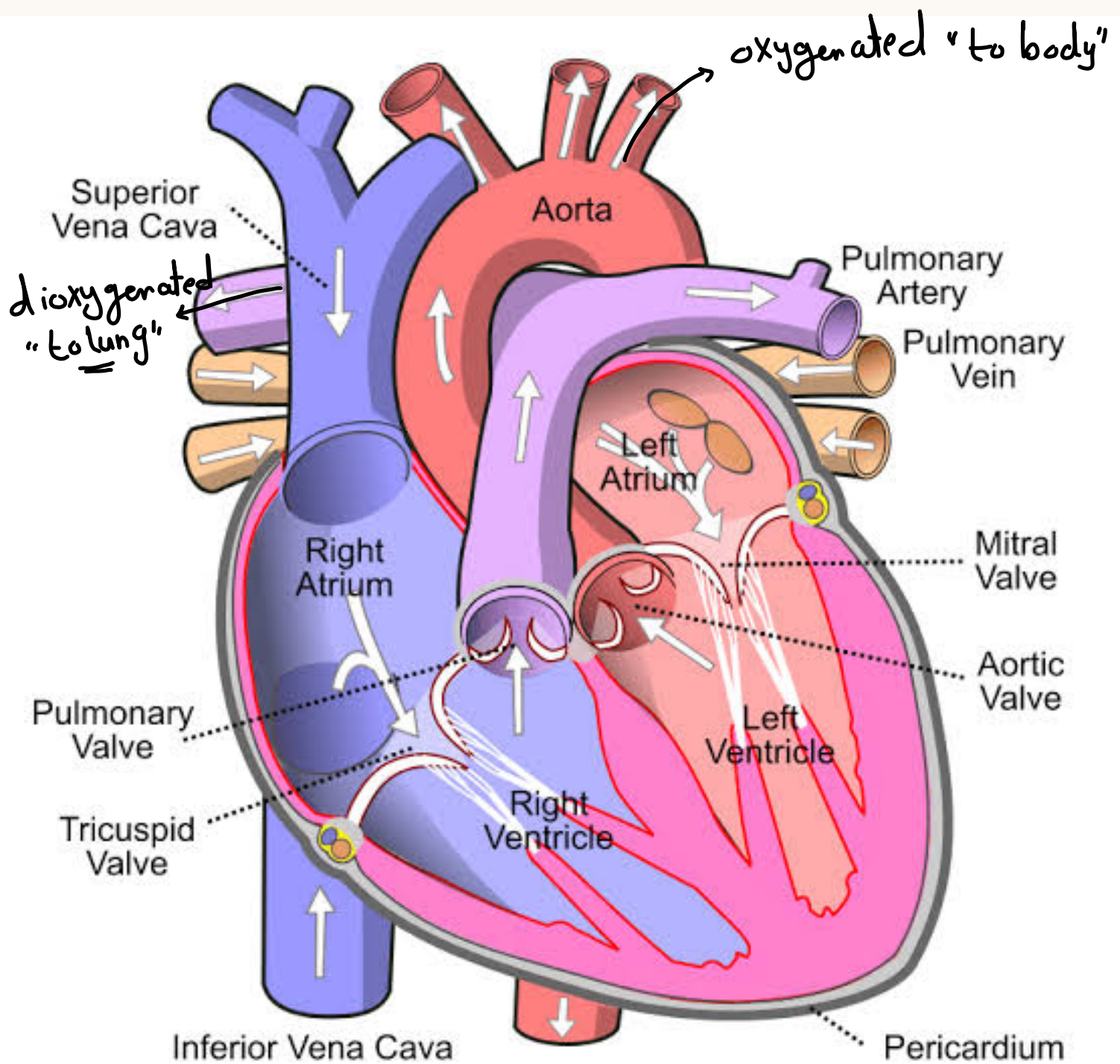
بعضی: کل لما مسافة الهواء اكن كل لما حدة الصوت
انقل. انخفاض
air harsh sound sharp sound
water

- ③ Sound Intensity $\propto \frac{1}{d^2}$
- Sound Intensity $\propto (\text{amplitude})^2$
 - Sound Intensity $\propto$ vibrating surface area.
 - Sound Intensity $\propto$ density of medium

Systems

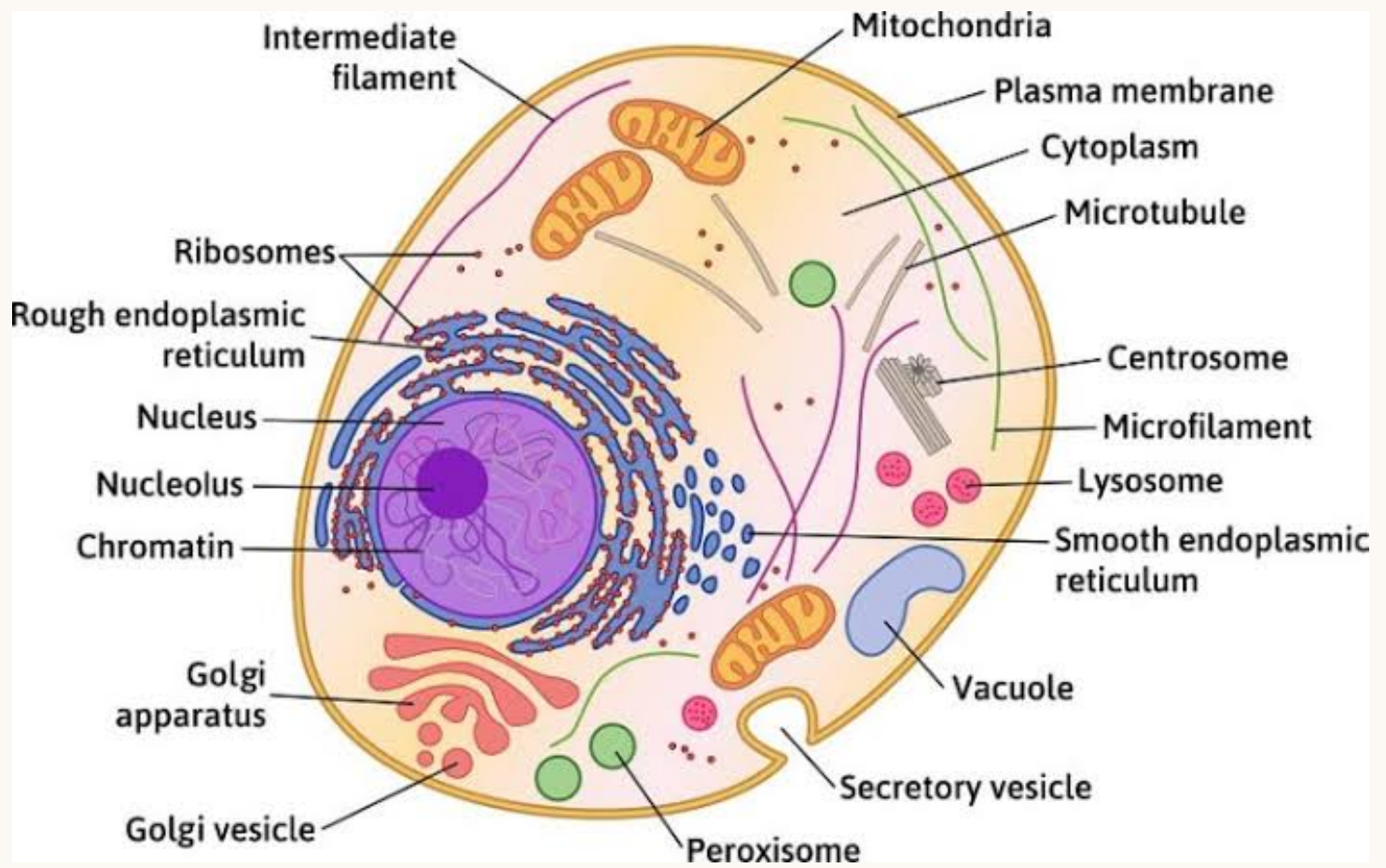






- deoxygenated
↳ pulmonary artery → blue
- oxygenated
↳ pulmonary vein → red

Animal cell



Plant Cell Structure

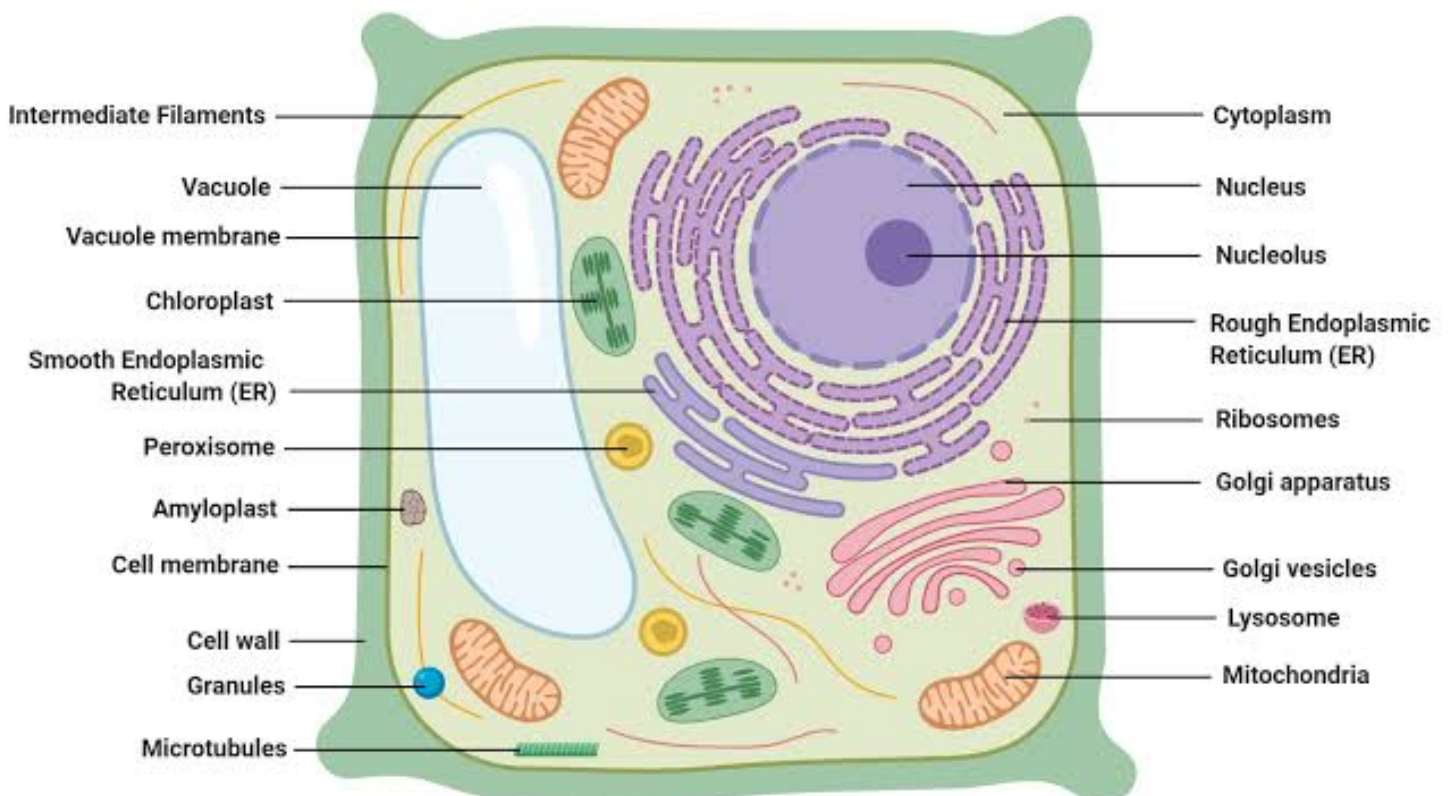


Figure: Plant Cell Structure, Image Copyright © Sagar Aryal, www.microbenotes.com

① Growth hormon

After puberty

• Achromegaly

↓ تضخم العظام

before puberty

"childhood"

↓ قزامة Dwarfism

↓ ضخامة Gigantism

② Thyroid gland

exophthalmic goiter

↓ Increase in:

- Metabolism
- heat
- under weight
- Sweating

Simple goiter

- Metabolism ↓
- cold ↑
- overweight ↑

③ Blood

RBCs

Transfer things all over the body

WBCs

Immunity

plasma

علاج العروق

platelets

& blood coagulators
"clotting"

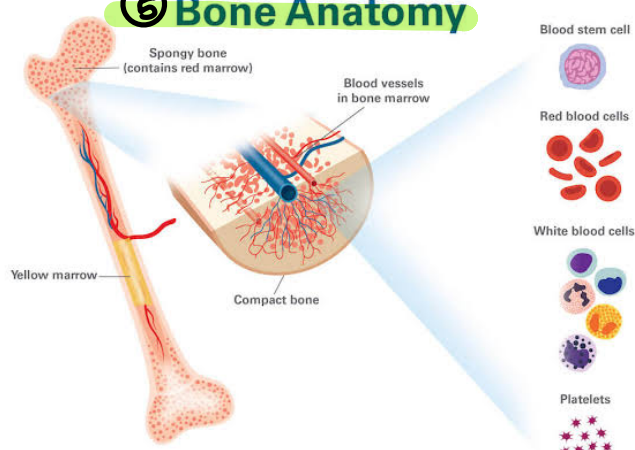
Hb → هيموجلوبين

protein حديد Fe

④ Types of muscles

	Main features	Location	Type of cells	Histology
Skeletal muscle	- Fibers : striated, tubular and multi nucleated - Voluntary - Usually attached to skeleton			
Smooth muscle	- Fibers : non-striated, spindle-shaped, and uninucleated. - Involuntary - Usually covering wall of internal organs.			
Cardiac muscle	- Fibers : striated, branched and uninucleated. - Involuntary - Only covering walls of the heart.			

⑤ Bone Anatomy



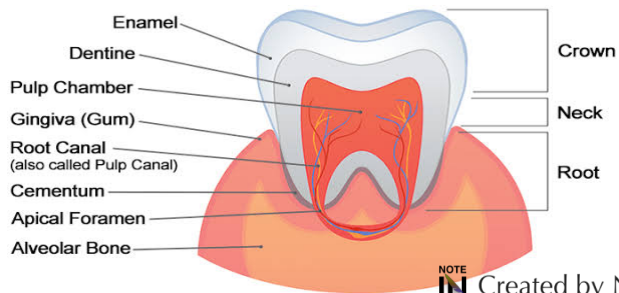
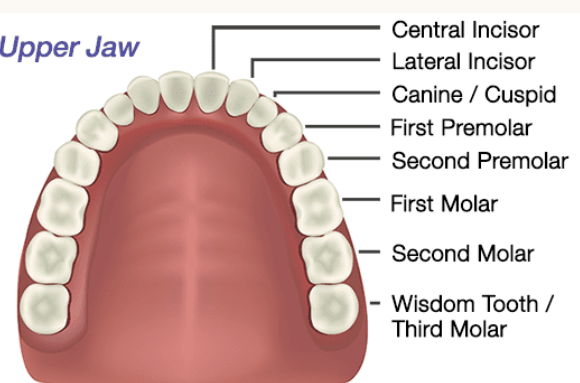
مسؤول عن تكوين الدم الجيد

Blood → Vein → Artery → بيتريش في الكلى Kidney

lymphs → in lymph vessels
go to lymph nodes → تتجمع من البكتيريا

parasympathic nerves	sympathic nerves.
قاع على الكنية heart beat ↓ breathe ↓	حنان Adrenaline ↑ heart beat ↑ breathe ↑

Upper Jaw



7

al

✓

Size increase

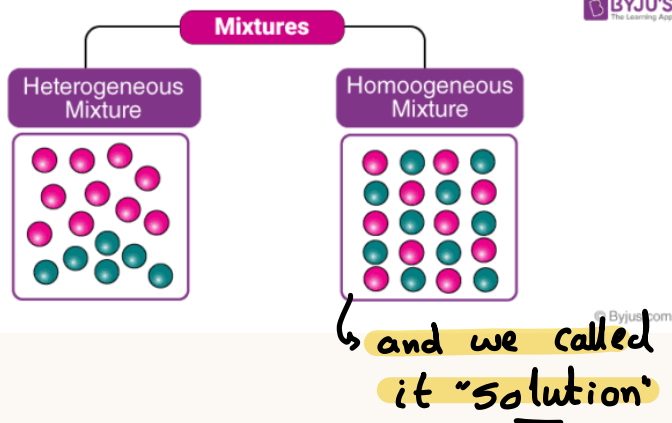
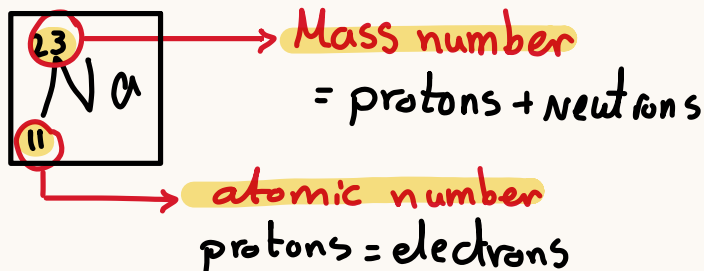
non-metallic activity decrease

strongest
non-metal

7

57	La	58	Ce	59	Pr	60	Nd	61	Pm	62	Sm	63	Eu	64	Gd	65	Tb	66	Dy	67	Ho	68	Er	69	Tm	70	Yb	71	Lu
Lanthanum [57La] 138.90486	Cerium [58Ce] 140.12577	Praseodymium [59Pr] 140.90765	Neodymium [60Nd] 144.242	Promethium [61Pm] 144.9127	Samarium [62Sm] 150.36	Europium [63Eu] 151.964	Gadolinium [64Gd] 157.25	Terbium [65Tb] 158.925	Dysprosium [66Dy] 162.50	Holmium [67Ho] 164.930	Erbium [68Er] 167.259	Thulium [69Tm] 168.930	Ytterbium [70Yb] 173.054	Lutetium [71Lu] 174.967															
67	Ac	72	Th	81	Pa	72	U	73	Np	74	Pu	75	Am	76	Cm	77	Bk	78	Cf	79	Es	80	Fm	81	Md	82	No	83	Lr
Actinium [89Ac] 227.02778	Thorium [90Th] 232.0377	Protactinium [91Pa] 231.03626	Uranium [92U] 238.02891	Neptunium [93Np] 237.04817	Plutonium [94Pu] 244.0642	Americium [95Am] 243.06138	Curium [96Cm] 247.07035	Berkelium [97Bk] 247.04637	Californium [98Cf] 251.07958	Einsteinium [99Es] 252.08321	Fermium [100Fm] 257.10528	Mendelevium [101Md] 258.10592	Nobelium [102No] 259.10858	Lawrencium [103Lr] 260.10539															

Compound $\rightarrow$ chemical changes
 $\hookrightarrow$ NaCl
 Solution $\rightarrow$ physical changes
 $\hookrightarrow$ NaCl $\cdot$ H_2O



Saturated Solutions



SOLUTION

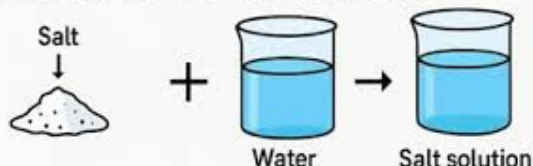
A solution is a homogeneous mixture of two or more substances.

Components of a Solution:

• **Solute:** The substance that dissolves (present in lesser quantity)

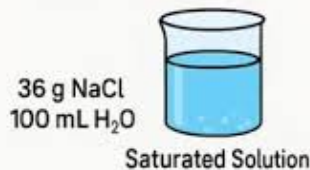
• **Solvent:** The substance in which solute dissolves (present in greater quantity)

Example: Salt (solute) + Water (solvent)



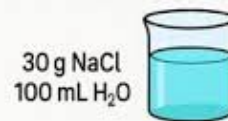
Saturated Solution

- No more solute can dissolve at a given temperature
- Extra solute settles at the bottom
- It is in equilibrium with undissolved solute

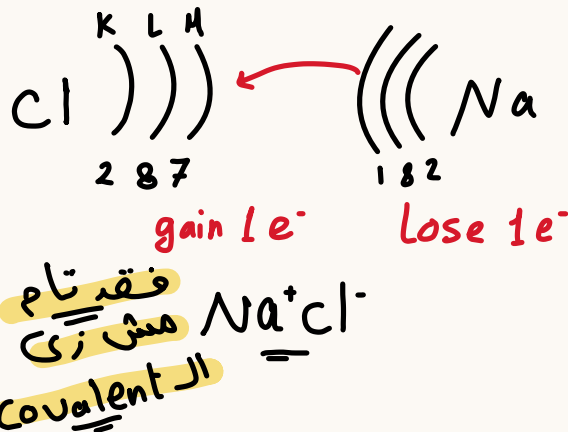


Unsaturated solution

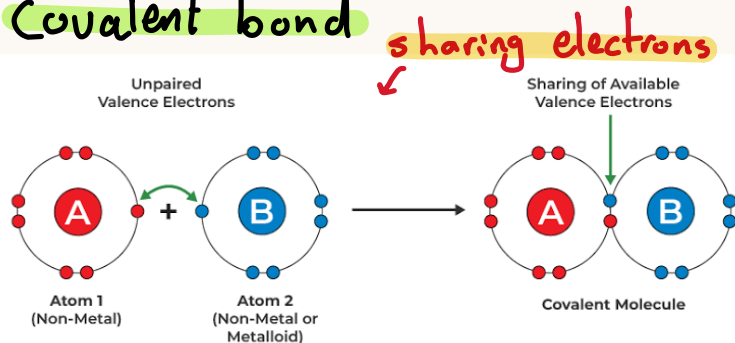
- Can still dissolve more solute at a given temperature
- No solute remains undissolved
- Not in equilibrium
Example: Water in which more salt can still dissolve



Ionic bond



Covalent bond



	No. of element	No. of Molecules	No. of atoms of Molecules
H_2O	2	1	3
2 NaCl	2	2	2
3MgCO ₃	3	3	5
C ₆ H ₁₂ O ₆	3	1	24

1. Combination or Synthesis Reaction



2. Decomposition Reaction



3. Single-replacement Reaction



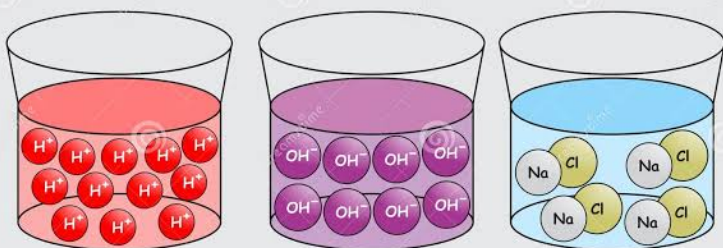
4. Double-replacement Reaction



5. Combustion Reaction



ACIDS, BASES AND SALTS



Acids

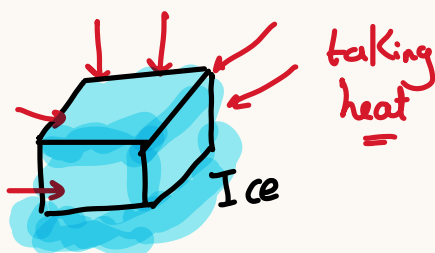
Donates a proton (H^+)
Sour tasting
Sticky in touch
In solution hydrogen ions rich
Turns blue litmus red

Bases

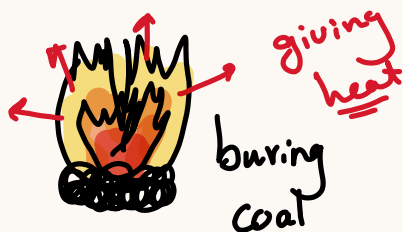
Accepts a proton (H^+)
Bitter tasting
Slippery in touch
In solution hydroxyl ions rich
Turns red litmus blue

Salts

Not classified as an acid or base
Dissolve in Water
Ionically Bond



"endothermic"



"exothermic"

the smallest degree of temperature is

$$\text{Zero } K \rightarrow -273^\circ C$$

morning

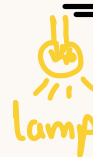


inside the house
"he can see outside"



outside
"he can see himself only"

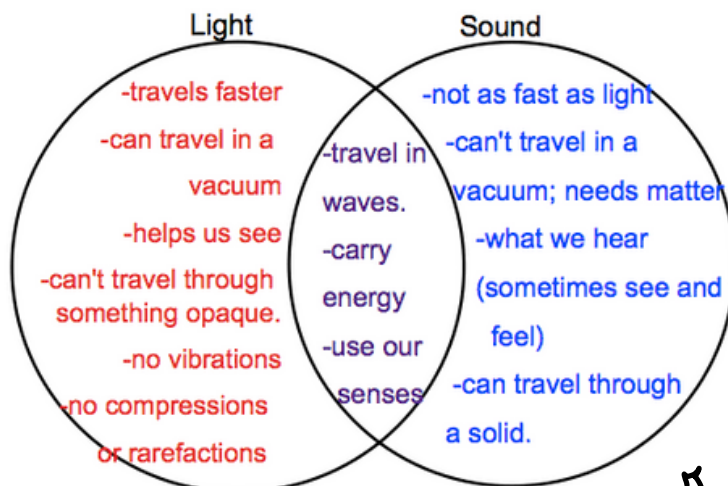
night



inside the house
"he can see himself only"

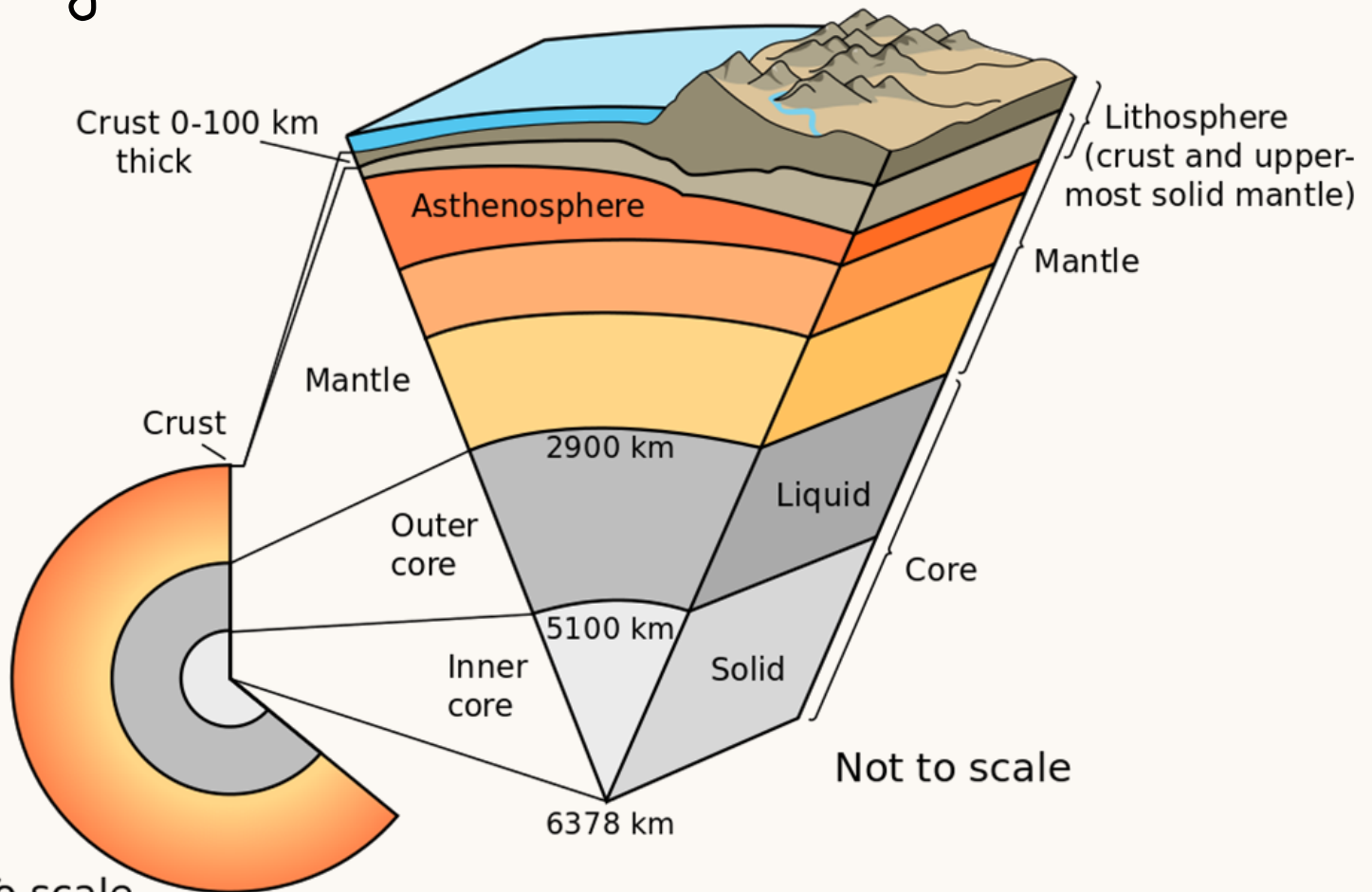


outside
"he can see inside"



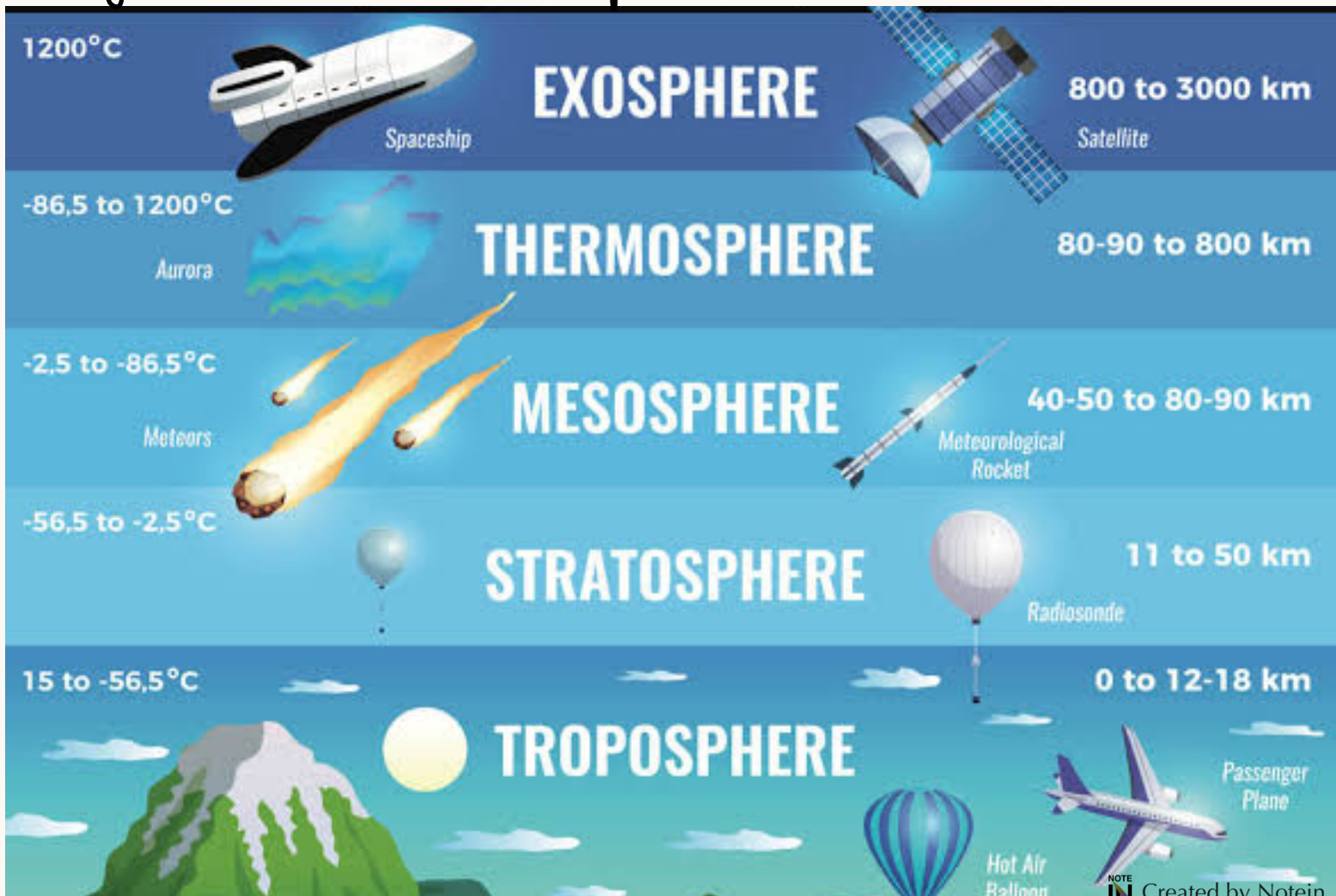
compare between Sound & light waves.

layers of the earth



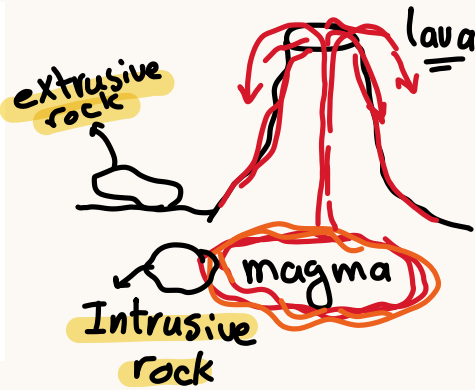
To scale

layers of the atmosphere



Types of rocks

① Igneous : "Ignis = fire"



② Sedimentary rocks:

رواسب = Sediments



minerals
+ small plants
+ organic
matters
= Sedimentary

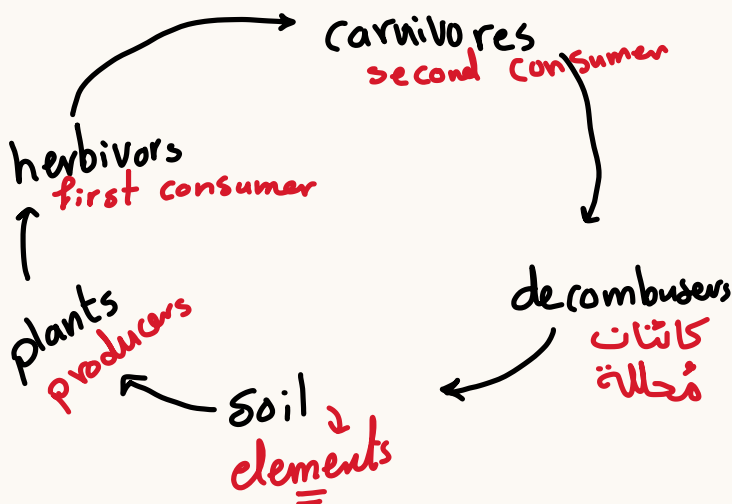
③ Metamorphic Rocks:



Sedimentary +
Igneous

exposed to
high Temp. and
pressure
for long time

Metamorphic



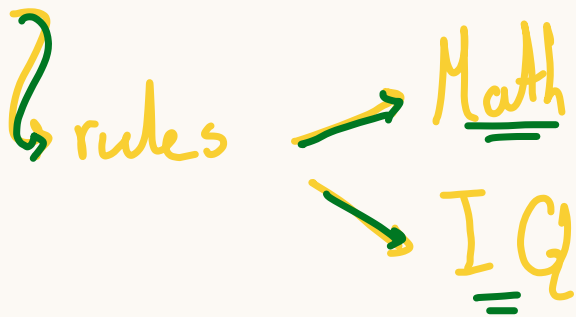
Mohs Hardness Scale

Mineral	Scale Number	Common Object
Diamond	10	Masonry Drill Bit
Sapphire	9	
Topaz	8	
Amethyst	7	
Opal	6	Steel Nail
Turquoise	5	
Pearl	4	Knife/Glass Plate
Coral	3	
Amber	2	Copper Penny
Talc	1	
		Fingernail

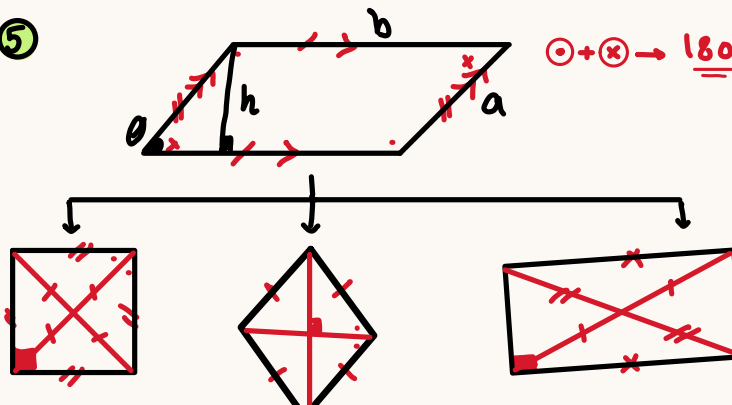
GABRIEL & CO.

↓
قوة يقدر يخدم الثاني
اعلى من الرقم يقدر يخدم الى اقل منه

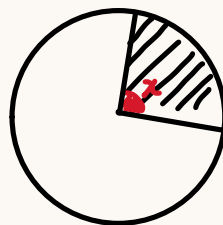
Mathematics



Rules

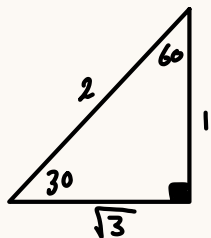
- ① Average = $\frac{\text{Sum}}{\text{number}}$
- ② Complete square $\rightarrow \sqrt[3]{\quad}$
 الناتج من غير decimals . e.g $\rightarrow \sqrt[3]{27} = 3$ ✓
 $\sqrt[3]{2^x} = 2^{\frac{x}{3}}$
- ③ Mean = Average
 Mode $\rightarrow$ التكرار
 Median $\rightarrow$ ترتيب بعد بين نصف النصف
- ④  $\odot + \otimes \rightarrow 180$
- ⑤ periodic number: $\frac{1}{3} = 0.\dot{3} = 0.3333\dots$
- ⑥ If two heights in a triangle are equal then the triangle $\rightarrow$ equilateral
- ⑦ Square: perimeter = side $\times 4$
 Area = (side)²
 Area = $\frac{1}{2} d^2$ $\rightarrow$ diagonal.
- ⑧ Rhombus: p = side $\times 4$
 Area = $\frac{1}{2} (d_1 \times d_2)$
- ⑨ parallelogram: p = $2(a+b)$
 $= 2a + \frac{2h}{\sin \theta}$
 Area = base $\times$ height

- ⑩ Rectangle: p = $2(l+w)$
 A = $L \times w$
- ⑪ Trapezium: $A = \frac{1}{2} (b_1 + b_2) h$
 middle base
- ⑫ cuboid: Lateral Area = perimeter of $\times$ height
 Total Area = (L.A) + 2 bases
 volume = length $\times$ w $\times$ h
- ⑬ cube: L.A = $S^2 \times 4$
 T.A = $S^2 \times 6$
 Volume = S^3
- ⑭ cylinder: T.A = $2\pi r(r+h)$
 L.A = $2\pi r \times h$
 volume = $\pi r^2 h$
- ⑮ Cone: L.A = $\pi r \times \text{length}$
 T.A = $\pi r(L+r)$
 Volume = $\frac{1}{3} \pi r^2 h$
- ⑯ sphere: surface area = $4\pi r^2$
 volume = $\frac{4}{3} \pi r^3$
- ⑰ Circumference of circle sector:
 = length of arc + 2r
 $(\frac{x}{360} \times 2\pi r)$

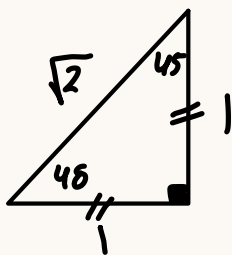


19) المثلث النذتين ستيني

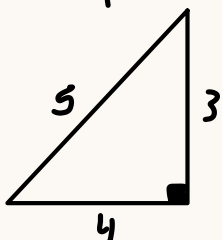
ratio $1 : \sqrt{3} : 2$



ratio $1 : 1 : \sqrt{2}$



ratio $3 : 4 : 5$



20) power = power, base $\neq$ base

$$\hookrightarrow 3^2 \times 5^2 = 15^2$$

power $\neq$ power, base = base

$$\hookrightarrow 3^7 \times 3^6 = 3^{13}$$

power = power, base = base

$$\hookrightarrow 7^6 \times 7^6 = 49^6$$

21) $\frac{\text{عدد النسلجات}}{2} = \frac{\text{العدد (العدد - 1)}}{2}$

22) Total of boys = $\frac{\text{Sum}}{(1 - \text{العدد})}$

23) angle = $\frac{(n-2) \times 180}{n}$ for each

24) angle of equalateral triangle:

$$= \frac{\sqrt{3}}{4} \times \text{side}^2$$

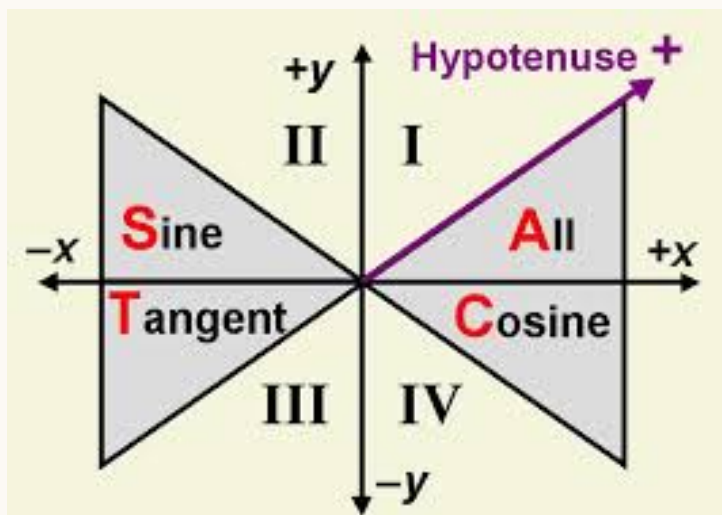
25) $-1 \leq \sin x \leq 1$

$-1 \leq \cos x \leq 1$

26) there is -ve angle and angles more than 360°

27) Slope of x-axis $\rightarrow$ zero

slope of y-axis $\rightarrow$ undifind.



28) how to find a slope:

$$\begin{array}{ccc} \text{①} & \text{②} & \text{③} \\ \frac{y_2 - y_1}{x_2 - x_1} & \tan \theta & y = mx + c \\ & \text{between } x\text{-axis} & \\ & \text{and line.} & \end{array}$$

29) $f(x) = 2x$

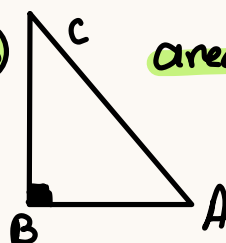
$$x = \{1, 2, 3\}$$

Domain

$$y = \{2, 4, 6\}$$

codomain

30) area of right triangle.



$$\frac{1}{2} \times \text{side} \times \text{side} = \frac{1}{2} \times c \times a \times \sin A$$

angle between $\underline{s_1}$ and $\underline{s_2}$

$$\hookrightarrow \frac{1}{2} \times c \times a \times \sin A$$

31) Distance between two points

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

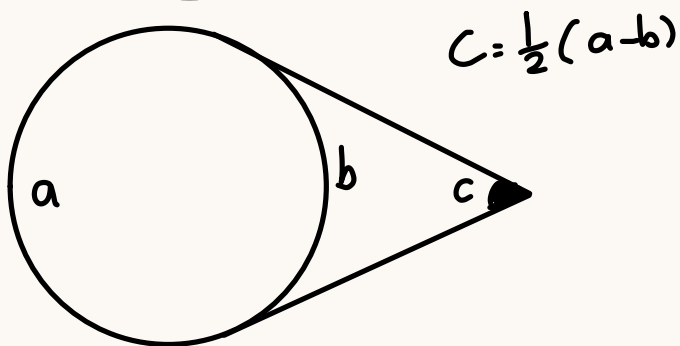
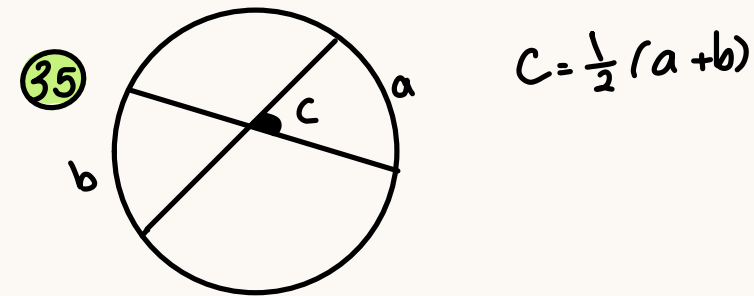
32) area of any regular shape
the number of its sides is "n"
and 'a' is length of the sides

$$\frac{n}{4} \times a^2 \times \tan\left(90 - \frac{180}{n}\right)$$

33) total = original amount $\times \left(\frac{100 + \text{ratio}}{100}\right)^{\text{years}}$

34) عدد ترتيب الأشياء على الرف
الرقم $\times$ بعين
"factorial" ←

eg → 6 objects → 6! → 6 × 5 × 4 × ...



36) angle of stars (sum)

$$= (N-4) \times 180$$

eg (5-4) × 180 →

